

Uncertainty-Aware Human Querying for Knowledge-Based Robot Planning

Changmin Park¹, Seongah Park¹, Nayoung Kim¹, and Sungjin Park¹
Geonyeong Ko¹, Junho Lee¹, and Changjoo Nam¹

¹Sogang University, Seoul, Republic of Korea

cmp9877@gmail.com

Abstract

Robots operating in real environments must act from incomplete symbolic knowledge while continuously incorporating uncertain perceptual observations. This paper presents a knowledge-based belief-space planning system that treats human querying as a belief refinement operation within the robot’s decision loop. The system partitions the knowledge base into domain, static, and local information; constructs an action-conditioned frontier belief over reachable symbolic successor states; updates particle weights after each observation; and computes entropy-normalized confidence before committing any state back to the knowledge base. When the belief is insufficiently confident, the robot selects the symbolic hypothesis expected to maximally reduce posterior entropy and queries the human through a feedback manager, rather than asking for action-level decisions or relying on continuous supervision. Online action selection is performed with a POMCP-style planner whose search is constrained by symbolic action applicability. We evaluate the resulting closed-loop system in waste sorting and tomato harvesting domains, including threshold analysis and policy comparisons against no-query, random-query, and always-query baselines. User-study results are reported as a secondary evaluation of interaction cost, while the main focus is the system-theoretic connection between symbolic knowledge, frontier belief update, confidence-triggered querying, and planning performance.

Keywords: belief-space planning, human-robot interaction, knowledge-based robotics, POMDP, proactive querying, planning under uncertainty

1 Introduction

Recent advances in robotic systems have improved autonomy in perception, manipulation, and planning, yet robots still fail in real-world environments due to perception errors, action execution failures, partial observability, and dynamic changes [22, 21, 26]. These failures often arise because the robot must select actions from an incomplete or ambiguous estimate of the current world state. In tasks such as tomato harvesting and waste sorting, uncertainty about object states or categories may remain even after sensing, making human feedback a useful source of task-relevant information.

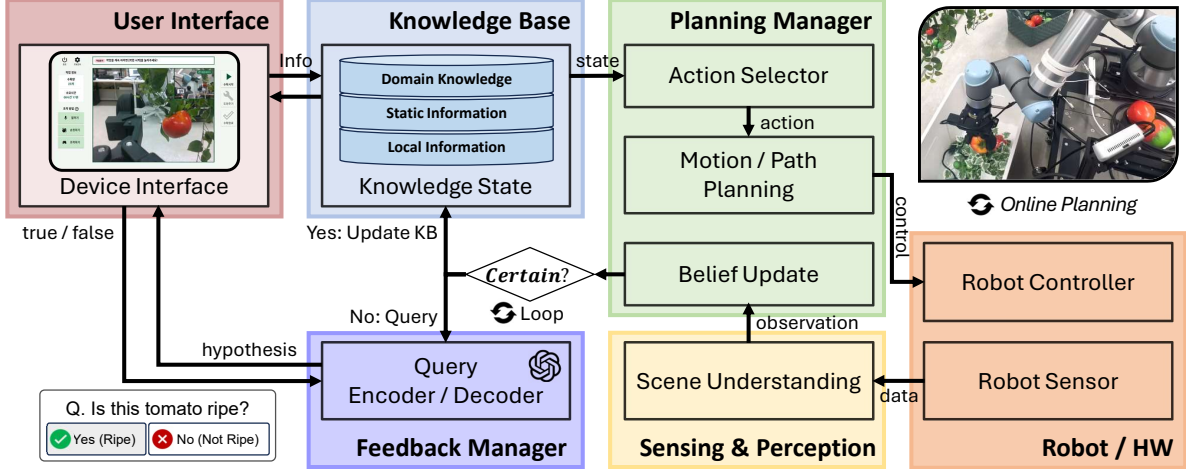


Figure 1: System architecture and data flow of the proposed framework. The knowledge base stores committed symbolic information, the planning manager maintains and updates the frontier belief, the sensing and perception module provides observations, and the feedback manager queries the user only when belief confidence is insufficient for knowledge-base commitment.

Existing ways of incorporating humans into robot autonomy remain limited. Continuous-supervision and human-on-the-loop systems rely on operators to monitor execution and detect failures [13, 10, 47, 39], which can increase workload and fatigue [4, 11, 55]. Failure-triggered systems reduce monitoring but ask only after errors occur, while fixed or uncertainty-threshold query systems can ask too often [44, 31, 38]. These approaches leave two coupled questions unresolved: *when* should the robot ask a human, and *what* should it ask?

We trigger human feedback from the robot’s belief update rather than treating it as continuous supervision or failure recovery. After each action, the robot compares the observation with possible transition outcomes encoded in its symbolic belief. If multiple task-relevant successor states remain plausible, the system estimates uncertainty using entropy and asks about the symbolic hypothesis expected to reduce the ambiguity. This follows the broader goal of selective assistance, where robots request human input only when it is useful [45, 3, 29].

This paper proposes a knowledge-based belief-space planning framework for proactive human querying. The robot maintains a symbolic knowledge base [8] with STRIPS-style state and action structure [12], while online decision making is performed using a POMDP formulation and POMCP-based planning [21, 49]. Instead of committing uncertain perceptual outputs as deterministic facts, the robot maintains a particle belief over symbolic possible worlds. After each observation, the system updates the belief over reachable frontier states, computes entropy-based confidence, and either commits the most likely state to the knowledge base or asks the human a targeted question before replanning.

The framework differs from continuous supervision, failure-triggered intervention, always-query policies, and action-level query systems: the user is not asked to monitor every decision or choose robot actions, but instead answers targeted questions about uncertain symbolic state facts.

We evaluate the framework in waste sorting and tomato harvesting. The main evaluation compares system performance under different confidence thresholds and query policies, while a com-

pact user study analysis examines whether state-level querying changes workload and situation-awareness measures.

The contributions and findings of this paper are:

- We formulate human querying as belief refinement in a symbolic planning system: observations update an action-conditioned frontier belief, entropy-normalized confidence determines whether the state can be committed, and queries are selected by expected posterior entropy.
- We present an integrated architecture connecting a structured knowledge base, POMCP-style online planning, sensing and perception, feedback management, and knowledge-base commitment.
- We show that the confidence threshold τ controls a sharp autonomy-interaction trade-off, with high task success emerging once the belief is sufficiently constrained by selective feedback.
- We compare no-query, random-query, always-query, and proposed query policies to isolate the value of belief-triggered, hypothesis-level querying; user-study results are included as a secondary analysis of interaction burden.

2 Related Work

In this section, we review prior work on knowledge-based robot planning, belief-space planning under uncertainty, human involvement in robot autonomy, and uncertainty-aware query systems.

2.1 Knowledge-Based Robot Planning

Knowledge-based robotic systems use structured representations to describe objects, properties, relations, actions, and task constraints. Logic-based knowledge systems and robot knowledge bases have been used to store observed facts, infer missing information, and support semantic reasoning in robot tasks [53, 33]. For example, robots can infer likely object locations from qualitative spatial relations, container-location relations, or place-object co-occurrence knowledge [25, 15]. Commonsense and relational reasoning further connect symbolic knowledge with perceptual evidence in partially observable environments [37, 58, 7]. Ontologies and type hierarchies support consistent integration of new observations and generalization across related objects, actions, and task contexts [36, 40, 35].

Classical symbolic planning provides a computational layer on top of such knowledge representations. STRIPS and PDDL represent states as sets of symbolic facts and actions as preconditions and effects [12, 1]. Related planning representations such as FDR and domain-independent planners expose compact state variables and enable efficient search over symbolic transition systems [17, 18, 16, 5]. This structure is useful in robotics because high-level actions can be checked against symbolic preconditions and then grounded into executable robot skills. However, classical symbolic planning typically assumes that the current state is known, whereas real robot knowledge is often produced by noisy perception and partial observations.

Recent work has expanded robot knowledge beyond hand-coded symbolic rules using large language models and vision-language models. LLM-based planning systems use procedural and

commonsense knowledge to propose task decompositions or action sequences, which are then grounded with robot affordances, programmatic interfaces, or execution feedback [2, 20, 50, 30, 9]. Other systems construct richer environment knowledge from scene graphs and open-vocabulary spatial representations, allowing robots to reason about objects, places, and relations beyond a closed symbolic vocabulary [14, 28]. Retrieval-augmented methods use demonstrations, procedural knowledge, or task-relevant constraints to guide long-horizon robot planning [51, 60, 57, 54, 24]. Related approaches combine LLM-generated plans with symbolic constraints, spatial reasoning, or verification mechanisms to improve executability and consistency [19, 6, 32, 42, 43].

These systems show that modern knowledge sources can greatly broaden what robots know and how they generate plans. Nevertheless, both traditional knowledge bases and foundation-model-based systems often commit inferred knowledge as a proposal, prior, or verified fact rather than maintaining an explicit execution-time belief over alternative symbolic worlds. Their outputs may remain uncertain because of perception errors, incomplete grounding, missing observations, or hallucinated semantic assumptions. Our work keeps the interpretability and action structure of symbolic planning while representing uncertain local facts as a belief over possible symbolic states. This lets the planner reason about what it does not know before committing perceptual outputs or inferred facts to the knowledge base.

2.2 Belief-Space Planning and Symbolic Uncertainty

POMDPs provide a standard framework for decision making under partial observability [21]. Instead of assuming access to the true state, a POMDP maintains a belief distribution over possible states and selects actions to maximize expected return. Online methods such as POMCP scale this idea by using Monte Carlo tree search and particle beliefs over action-observation histories [49]. Related online POMDP methods have been applied to robot navigation, object search, manipulation, exploration, and interaction under uncertainty [52, 27, 56, 41].

These methods show that particle beliefs are effective when the robot must preserve multiple hypotheses about the world. However, many POMDP formulations operate over task-specific state variables rather than explicit symbolic knowledge bases. Conversely, symbolic planners expose action preconditions and logical structure, but often lack a principled belief representation. Prior work on symbolic planning under uncertainty addresses this gap by combining logical structure with probabilistic inference or POMDP-style planning [46, 59, 48]. Such methods demonstrate that symbolic structure can help organize uncertainty, but uncertainty is usually resolved through autonomous sensing, transition models, or planning abstraction.

Our setting adds a different source of information: human semantic feedback. Some ambiguities cannot be reliably resolved by additional robot sensing alone, such as visually ambiguous object states or uncertain task-relevant categories. We therefore maintain a symbolic particle belief over reachable frontier states and allow the robot to query a human when the belief is insufficiently confident. The answer directly filters the planner’s belief and updates the knowledge base.

2.3 Human Involvement in Robot Autonomy

Human involvement has long been used to improve robot reliability under uncertainty. In teleoperation and supervisory control, humans monitor the robot and intervene when needed [13, 10]. Human-on-the-loop systems preserve robot autonomy during nominal operation while relying on

an operator to detect and correct failures [47, 39, 23, 34]. These systems can improve safety and robustness, but they require sustained human attention. The operator must notice that a problem exists, interpret the situation, and decide how to intervene, which can increase workload and reduce scalability in long-horizon or repetitive tasks [11, 55].

Another approach is failure-triggered assistance, where the robot asks for help only after it fails, reaches a dead end, or detects an invalid state. This reduces continuous monitoring, but it treats human input as recovery rather than prevention. In tasks with irreversible or costly actions, asking after failure may be too late. A third approach is frequent or fixed querying, where the robot asks whenever a predefined uncertainty condition is met. This can improve task success, but it often produces unnecessary interactions because it does not distinguish uncertainty that matters for the current plan from uncertainty that is irrelevant.

Our work targets a proactive middle ground. The human is not asked to continuously supervise the robot, and the robot does not wait until failure. Instead, the robot estimates whether its current symbolic belief is confident enough to support planning. If not, it asks a targeted question about the symbolic fact expected to reduce the most uncertainty.

2.4 Uncertainty-Aware Querying

Recent work has studied robots and embodied agents that ask for help based on uncertainty. Active learning and active preference learning query humans to improve models or infer preferences [45]. Proactive robot behavior has also been shown to improve collaboration when the robot can anticipate when assistance is useful [3]. More recent uncertainty-aware systems use model confidence, language-model uncertainty, or conformal prediction to decide when to request human input [44, 31, 38]. These methods are important because they move beyond continuous supervision and show that selective querying can reduce unnecessary intervention.

However, many query systems focus on action-level uncertainty or model-output confidence. In such systems, the human may be asked which action the robot should take, whether a proposed action is acceptable, or whether a model prediction should be trusted. This differs from asking about the latent symbolic state that causes planning uncertainty. If the robot asks for an action, the human effectively participates in planning. If the robot asks for a state fact, the human provides information while the robot remains responsible for planning.

Our framework follows the second approach. The robot asks predicate-level questions about task-relevant symbolic facts, such as object properties, perceived categories, or local symbolic conditions. Human responses eliminate inconsistent hypotheses from the belief and update the knowledge base, after which planning proceeds autonomously. By integrating selective querying with symbolic belief maintenance and online planning, our approach treats human feedback as an information source for reducing planning uncertainty rather than as a mechanism for action selection.

3 Preliminaries

3.1 Symbolic Planning

Let $\mathcal{F}$ be a finite set of symbolic atoms. A symbolic state is a subset $s \subseteq \mathcal{F}$. An action $a \in \mathcal{A}$ is defined by preconditions, add effects, and delete effects:

$$a = \langle \text{pre}(a), \text{add}(a), \text{del}(a) \rangle.$$

The action is applicable in state s if $\text{pre}(a) \subseteq s$. Applying a produces

$$s' = (s \setminus \text{del}(a)) \cup \text{add}(a).$$

In robot domains, symbolic actions are grounded to executable skills such as detecting an object, navigating to a location, picking an object, placing an object, scanning a tomato, or discarding an item.

3.2 POMDPs and POMCP

A POMDP is a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, T, R, \Omega, O, \gamma \rangle$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $T(s'|s, a)$ the transition model, $R(s, a)$ the reward function, Ω the observation space, $O(o|s', a)$ the observation model, and γ the discount factor. Since the state is not directly observed, the robot maintains a belief $b(s)$ over states. After action a and observation o , the Bayesian belief update is

$$b'(s') = \eta O(o|s', a) \sum_{s \in \mathcal{S}} T(s'|s, a) b(s), \quad (1)$$

where η is a normalizer.

POMCP approximates online POMDP planning through Monte Carlo tree search over histories [49]. Each history node stores sampled particles and action values. During simulation, actions are selected using an upper-confidence rule,

$$a^* = \arg \max_a \left[V(ha) + c \sqrt{\frac{\log N(h)}{N(ha)}} \right], \quad (2)$$

where $V(ha)$ is the estimated value of taking action a after history h , $N(h)$ is the visit count of the history, and $N(ha)$ is the visit count of the action branch.

3.3 Human Feedback as Information

We treat a human response as an observation over a symbolic hypothesis. A hypothesis is a literal $q \in \mathcal{F}$ whose truth value is uncertain in the current belief. A query asks the human to verify q , producing an answer $y \in \{\text{true}, \text{false}\}$. The answer filters the belief by retaining states consistent with y . Thus, a query affects future planning by changing the posterior over symbolic worlds.

4 Method

4.1 Problem Formulation

The robot is given a symbolic domain, a set of executable skills, an initial knowledge state, and a task goal. Some facts are known with certainty, while local task facts must be inferred from observations. The robot selects task actions $a \in \mathcal{A}$ using POMCP. After each action, the observation may support several possible successor states rather than a single deterministic outcome. The system then computes belief uncertainty and requests human feedback only if the updated belief is not confident enough to commit a symbolic state to the knowledge base. The objective is to maximize task reward while reducing unnecessary feedback requests:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t (R(s_t, a_t) - \lambda \mathbb{I}[\text{query}_t]) \right], \quad (3)$$

where query_t indicates that feedback is requested after the belief update at time t , and λ is the cost of human interaction.

The key system-level distinction is between the *committed knowledge state* and the *belief state*. The committed state s_t^K contains facts that the robot is willing to use as knowledge for symbolic planning and skill execution. The belief $\mathcal{B}_t$ is a temporary distribution over candidate successor states induced by recent actions and observations. Human feedback is invoked only between these two layers: it refines $\mathcal{B}_t$ before the system commits a new s_{t+1}^K . Thus, querying is not modeled as a separate recovery mode, but as an information action in the knowledge-update loop.

4.2 Knowledge Base

The knowledge base is partitioned into domain information, static information, and local information:

$$\mathcal{KB} = \mathcal{I}^D \sqcup \mathcal{I}^S \sqcup \mathcal{I}^L. \quad (4)$$

Domain information encodes predicates, action schemas, task rules, and robot skills. Static information contains facts and numeric fluents that remain fixed during an episode, such as object identifiers, bin locations, or stem locations. Local information contains facts inferred during execution, such as object category, tomato state, object pose, and visibility. Uncertainty is concentrated in $\mathcal{I}^L$ because these facts depend on perception and partial observations. Each symbolic state is therefore represented as

$$s^K = \mathcal{I}^D \cup \mathcal{I}^S \cup \hat{\mathcal{I}}^L, \quad (5)$$

where $\hat{\mathcal{I}}^L$ denotes the subset of local information that has been committed after belief update and, when needed, human feedback. This separation prevents uncertain perceptual outputs from being immediately treated as certain planning facts.

4.3 Symbolic Frontier Belief

Let s^K denote the current committed knowledge state. Instead of constructing a global belief over all symbolic states, we construct a local belief over states reachable from s^K under the selected

task action. For action a , the symbolic frontier is

$$\text{Frontier}(s^K, a) = \{s_1^*, \dots, s_j^*\}, \quad (6)$$

where each s_k^* is a reachable successor hypothesis consistent with the action model and the current knowledge base. The belief over the frontier is represented by weighted particles:

$$\mathcal{B} = \{(s_k^*, w_k)\}_{k=1}^j, \quad \sum_{k=1}^j w_k = 1. \quad (7)$$

After executing a and receiving observation o , each particle weight is updated by the observation model:

$$w'_k = w_k O(o|s_k^*, a), \quad \tilde{w}_k = \frac{w'_k}{\sum_{\ell=1}^j w'_\ell}. \quad (8)$$

This local frontier construction has three roles. First, it avoids enumerating irrelevant symbolic worlds. Second, it preserves symbolic action constraints because every particle is reachable under the current action model. Third, it ties uncertainty to the action currently affecting task progress, which makes later human queries task-relevant. The resulting belief is local in scope but sufficient for deciding whether the knowledge base can be safely updated after the current transition.

4.4 Confidence and Query Trigger

We quantify uncertainty using the entropy of the normalized frontier belief:

$$H(\mathcal{B}) = - \sum_{k=1}^j \tilde{w}_k \log_2 \tilde{w}_k. \quad (9)$$

The confidence score is

$$\text{Conf}(\mathcal{B}) = \begin{cases} 1, & j = 1, \\ 1 - \frac{H(\mathcal{B})}{\log_2 j}, & j > 1. \end{cases} \quad (10)$$

If $\text{Conf}(\mathcal{B}) \geq \tau$, the robot commits the maximum-weight state to the knowledge base:

$$s^K \leftarrow \arg \max_{s_k^* \in \mathcal{B}} \tilde{w}_k. \quad (11)$$

If $\text{Conf}(\mathcal{B}) < \tau$, the robot requests human feedback. The threshold τ is therefore an autonomy-interaction parameter: lower values allow the robot to act with less certainty, while higher values increase interaction to obtain more reliable knowledge.

The confidence rule defines a guarded transition from belief to knowledge. When confidence is high, the update

$$(s_t^K, \mathcal{B}_t, o_{t+1}) \mapsto s_{t+1}^K \quad (12)$$

is performed autonomously by committing Eq. 11. When confidence is low, the same transition is delayed until one or more feedback answers filter the belief. This guard is the mechanism that connects uncertainty estimation to system behavior.

4.5 Query Selection

The system constructs candidate hypotheses from literals whose truth values differ across the frontier states. Let $\mathcal{C}$ be this set. For each candidate $c \in \mathcal{C}$, the belief is partitioned into states where c is true and false:

$$\mathcal{B}_c^+ = \{(s_k^*, \tilde{w}_k) \mid c \in s_k^*\}, \quad \mathcal{B}_c^- = \{(s_k^*, \tilde{w}_k) \mid c \notin s_k^*\}.$$

The selected query minimizes expected posterior entropy:

$$c^* = \arg \min_{c \in \mathcal{C}} [P(c)H(\mathcal{B}_c^+) + P(\neg c)H(\mathcal{B}_c^-)], \quad (13)$$

where $P(c) = \sum_{s_k^*: c \in s_k^*} \tilde{w}_k$ and $P(\neg c) = 1 - P(c)$. After the human answers, the belief is filtered:

$$\mathcal{B} \leftarrow \begin{cases} \mathcal{B}_{c^*}^+, & y = \text{true}, \\ \mathcal{B}_{c^*}^-, & y = \text{false}. \end{cases} \quad (14)$$

The remaining weights are renormalized. Querying repeats until the confidence exceeds τ or no informative hypothesis remains.

4.6 Planning and Execution Pipeline

At each step, the planning manager runs POMCP using the current belief and the symbolic action constraints, following the closed-loop architecture in Fig. 1. The selected task action is executed by a corresponding robot skill. Perception produces observations used to update the frontier belief. If the updated confidence is low, the feedback manager converts the selected symbolic hypothesis into a human-readable query, highlights the relevant object in the interface, receives a binary response, and returns the response to the planner. The knowledge base is updated only after the belief is sufficiently confident.

Action selection follows the POMCP structure, but the search is restricted to actions applicable in the sampled symbolic state. This prevents the planner from evaluating infeasible symbolic transitions and keeps Monte Carlo simulations consistent with the knowledge base. In this sense, the system combines two forms of structure: symbolic preconditions constrain the search space, while frontier particles represent the remaining uncertainty after action execution.

5 Experiments

The experiments are organized around the belief-update mechanism rather than around user interaction alone. We first evaluate how the confidence threshold τ affects task success, execution length, and query frequency. This analysis directly tests the role of entropy-normalized confidence in deciding when belief can be committed to the knowledge base. We then compare query policies to isolate the value of belief-triggered, hypothesis-level querying. A compact user study is included as a secondary analysis of the human-facing consequences of these policies.

Input: initial knowledge state s_0^K , threshold τ
Output: executed action sequence
 $s^K \leftarrow s_0^K$;
for $t = 0$ **to** T **do**
 $a_t \leftarrow \text{POMCP}(s^K, \mathcal{B})$;
 if $a_t = \emptyset$ **then**
 | **break**
 end
 execute a_t and receive observation o_{t+1} ;
 construct $\text{Frontier}(s^K, a_t)$ and update $\mathcal{B}$ using Eq. 8;
 compute $\text{Conf}(\mathcal{B})$ using Eq. 10;
 while $\text{Conf}(\mathcal{B}) < \tau$ **do**
 | select c^* using Eq. 13;
 | query human for c^* and receive y ;
 | filter $\mathcal{B}$ using Eq. 14;
 | recompute $\text{Conf}(\mathcal{B})$;
 end
 commit the most likely state to s^K using Eq. 11;
end

Algorithm 1: Uncertainty-Aware Planning with Human Querying

5.1 Domains

Waste sorting. The waste sorting domain consists of a UR5 robot sorting objects into category-specific bins. Each object belongs to one of four categories: general waste, paper, can, or plastic. Categories are not fully known in advance and may be ambiguous due to occlusion or perception uncertainty. The robot must detect, pick, and place objects into the correct bins.

Tomato harvesting. The tomato harvesting domain contains tomatoes attached to stems. Each tomato may be ripe, unripe, or rotten. Ripe tomatoes should be harvested, rotten tomatoes should be discarded, and unripe tomatoes should be left untouched. This domain is more sensitive to unresolved local information because a wrong belief about a tomato state can directly cause a wrong harvest or discard.

5.2 Metrics and Policies

We report task success rate, average number of queries, and average number of execution steps. Query count measures interaction cost, while execution steps measure the downstream planning cost of unresolved belief ambiguity. In the policy comparison, **All** queries whenever feedback can be generated, **No** never queries, **Random** queries without using belief confidence, and **Ours** queries only when $\text{Conf}(\mathcal{B}) < \tau$ and selects the hypothesis minimizing expected posterior entropy.



(a) Waste sorting.



(b) Tomato harvesting.

Figure 2: Experimental domains. The waste sorting domain evaluates symbolic category uncertainty in simulation, while the tomato harvesting domain evaluates uncertainty over tomato states in a robot harvesting task.

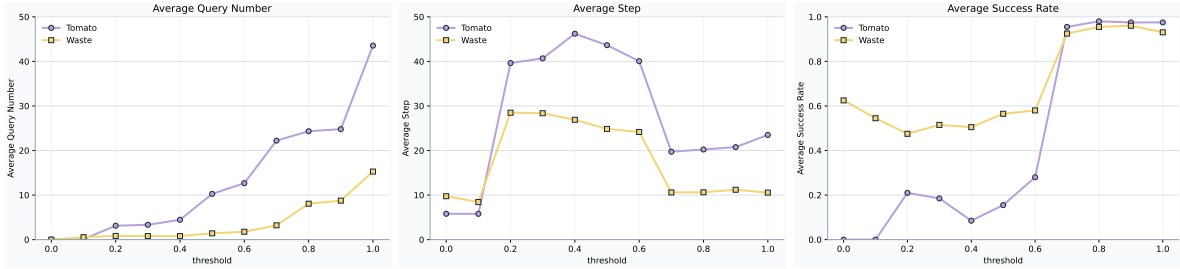


Figure 3: System performance under different confidence thresholds τ . Increasing τ makes the belief-to-knowledge transition more conservative, increasing query frequency while reducing failures caused by premature commitment.

5.3 Confidence Threshold Analysis

The threshold analysis shows that τ acts as a system-level guard on knowledge-base commitment. Low thresholds allow the robot to commit states despite high entropy, which reduces queries but often leaves task-relevant ambiguity unresolved. In tomato harvesting, success remains below 30% through $\tau = 0.6$ because the robot frequently commits incorrect local information about tomato states. Increasing τ from 0.6 to 0.7 raises success from 28.0% to 95.5% in tomato harvesting and from 58.0% to 92.5% in waste sorting. This sharp transition indicates that belief confidence is not merely a descriptive statistic; it directly controls whether the symbolic planner acts from stable knowledge or from ambiguous local information.

For $\tau \geq 0.7$, both domains maintain success above 92%. We use $\tau = 0.8$ for policy comparisons because it gives high success in both domains while avoiding the excessive query cost observed at $\tau = 1.0$, especially in tomato harvesting.

5.4 Query Policy Comparison

The policy comparison separates the effect of feedback quantity from the effect of belief-aware timing. In waste sorting, **All**, **Random**, and **Ours** all reach 100.0% success, but **Ours** reduces queries relative to **All**. In tomato harvesting, where local state ambiguity is more consequential, **Ours** reaches 98.0% success with 20.5 queries on average, while **All** reaches 96.5% success with 37.3 queries. **No** fails frequently because incorrect beliefs are committed without correction, and

Table 1: System evaluation under different confidence thresholds τ . Success rates are percentages.

τ	Waste			Tomato		
	Success	Queries	Steps	Success	Queries	Steps
0.0	62.5	0.0	9.7	0.0	0.0	5.8
0.3	51.5	0.8	28.4	18.5	3.3	40.7
0.6	58.0	1.8	24.2	28.0	12.7	40.1
0.7	92.5	3.2	10.6	95.5	22.2	19.7
0.8	95.5	8.1	10.6	98.0	24.3	20.2
0.9	96.0	8.8	11.2	97.5	24.8	20.8
1.0	93.1	15.3	10.6	97.5	43.5	23.5

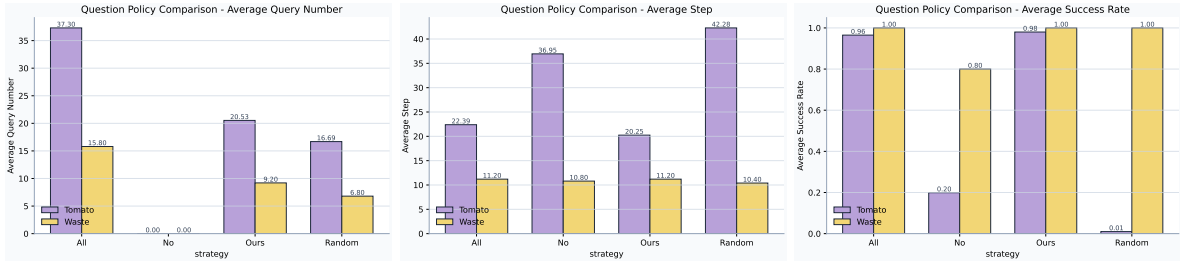


Figure 4: System performance under different query policies at $\tau = 0.8$. Belief-triggered querying preserves high success while avoiding the interaction cost of always querying.

Random performs poorly because queries are not aligned with the belief states that matter for future planning.

These results support the central system claim: the benefit comes from coupling the query trigger and query content to the frontier belief. Asking more often is not sufficient, and asking less often is not safe unless the system can determine when uncertainty matters for planning.

5.5 User Study as Secondary Evaluation

The user study is included to check whether the system-level query policy creates reasonable interaction demands for human users. Twelve participants completed waste sorting and tomato harvesting scenarios under five interaction conditions: **All**, **No**, **Ours1**, **Ours2**, and **KnowNo**. Because the IJRR framing focuses on the belief-space system, we summarize only the main interaction findings here.

Across both domains, SAGAT-based answer accuracy did not show a significant overall condition effect, but waste sorting showed a significant condition effect: **Ours1** achieved the highest accuracy ($M = 0.72$, $SD = 0.15$) and exceeded **All** ($M = 0.56$, $SD = 0.19$). The strongest item-level signal came from the question asking why the robot requested information; in waste sorting, **Ours1** reached perfect accuracy on this item. For workload, **Ours2** reduced NASA-TLX workload relative to the action-level **KnowNo** baseline and showed lower effort than both **All** and **KnowNo**. These results are consistent with the system view that state-level queries can expose the robot’s uncertainty without asking users to take over action selection.

Table 2: Query policy comparison at $\tau = 0.8$.

Policy	Waste			Tomato		
	Success	Queries	Steps	Success	Queries	Steps
All	100.0	15.8	11.2	96.5	37.3	22.4
No	80.0	0.0	10.8	20.0	0.0	36.9
Random	100.0	6.8	10.4	1.0	16.7	42.3
Ours	100.0	9.2	11.2	98.0	20.5	20.2

6 Discussion

The central claim of this paper is that asking a human should be treated as part of the robot’s belief dynamics, not as an external recovery behavior. This matters because the usefulness of a query depends on both the robot’s current posterior and the task consequences of committing the wrong symbolic state. A query about a visually ambiguous tomato is valuable if the answer changes whether the robot harvests, discards, or ignores it. The same query may be unnecessary if the tomato is irrelevant to the current goal or if another action can resolve the uncertainty autonomously.

The frontier belief representation makes this connection explicit. By constructing uncertainty around states reachable from the current knowledge state and selected action, the robot avoids asking about facts that do not affect near-term planning. The entropy-based confidence score determines when the robot lacks enough information to commit a state, and the expected-entropy query rule determines which symbolic fact would most reduce the ambiguity. In this formulation, the knowledge base is not simply a database of perceptual outputs. It is a guarded state estimator: local perceptual facts become planning knowledge only after the belief is sufficiently concentrated.

The system evaluation supports this interpretation. The sharp performance change around $\tau = 0.7$ suggests that the confidence threshold functions as a meaningful guard between ambiguous belief and committed knowledge. The policy comparison further shows that query timing cannot be replaced by query quantity alone: random feedback can spend interactions without stabilizing the belief states that matter for planning, while always querying incurs unnecessary interaction cost.

The user study should be interpreted as secondary evidence rather than the core contribution. Its results suggest that state-level queries can make the robot’s uncertainty more legible and can reduce workload relative to action-level query baselines. However, the paper’s main claim is not that one interaction condition universally improves all human factors measures. Rather, the human results are consistent with the system design principle that people should refine uncertain state information while the robot remains responsible for belief update and action selection.

Several limitations remain. First, the framework assumes that human answers are reliable or at least more reliable than the robot’s uncertain observation. Future work should model noisy human feedback as another observation source rather than as a hard filter. Second, the current query form is binary and predicate-level. More expressive questions may reduce the number of interactions but increase user burden. Third, the planner uses a local frontier belief rather than a full global posterior. This improves tractability and preserves symbolic action relevance, but it may miss dependencies outside the immediate action-conditioned frontier. Finally, the current implementation

relies on domain-specific frontier construction; learning or automatically generating these frontier models would make the framework easier to transfer across tasks.

7 Conclusion

We presented a knowledge-based belief-space planning framework that integrates proactive human querying into the robot’s state-estimation and planning loop. The robot maintains a committed symbolic knowledge state, constructs an action-conditioned frontier belief after execution, updates particle weights from observations, and uses entropy-normalized confidence to decide whether the belief can be committed back to the knowledge base. When confidence is insufficient, the system asks a targeted state-level question selected by expected posterior entropy. This formulation treats human feedback as belief refinement rather than action-level control. Across waste sorting and tomato harvesting, the results show that belief-triggered querying can preserve high task success while avoiding the interaction cost of always querying. The user study provides secondary evidence that state-level querying can support awareness and reduce perceived burden relative to action-level interaction baselines.

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A Appendix

A.1 Experiment Settings

This appendix should list the exact planner parameters, perception parameters, domain parameters, and scenario descriptions used in the waste sorting and tomato harvesting experiments.

A.2 Additional Threshold Results

Additional per-scenario threshold plots and raw statistics for τ should be placed here after the main threshold analysis is summarized in Sec. 5.